

GPR Shock Extraction and Baseline Local Projection

Summary Report

May 18, 2026

1 Overview

This report summarizes the construction of **standardized geopolitical risk (GPR) shocks** and their use as impulses in a **baseline local projection (LP) model** to estimate the dynamic response of oil-market prices and aggregate macro outcomes.

The pipeline consists of three stages:

- (i) **AR-residual extraction of standardized GPR shocks** z_t for three GPR series: LGPR (World), LGPRT (threats), LGPRA (acts). Two control specifications:

Regressor block	Timing	w/ oil	w/o oil
GPR own lags	$t - 1, \dots, t - 4$	✓	✓
Unemp, FFR	contemporaneous (t)	✓	✓
Real_Brent, Real_WTI, Real_Gasoline	lag-1 ($t - 1$)	✓	—

- (ii) **Three diagnostic tests on z_t to verify identification:**

#	Test	Tool	Lag(s)	N_{tests}
1	Conditional Granger exogeneity	gctest	4	30
2	Ljung-Box Q white noise	lbqtest	{4, 8, 12}	18
3	Breusch-Godfrey LM (robust)	aux. regression	{4, 8, 12}	18

N_{tests} : total tests run. Test 1 = 2 variants $\times$ 3 shocks $\times$ 5 predictors = 30; Tests 2–3 = $2 \times 3 \times 3$ lag horizons = 18 each.

- (iii) **Local projection on z_t for six outcome variables.** Settings: $H = 12$ quarters, 90% Newey–West HAC bands, bandwidth $h + 1$.

Block	Variables
Oil prices	Real_Brent, Real_WTI, Real_Gasoline
Macro	Headline_Pi, Core_Pi, Indu_Prod

2 Data

(1) Source

- File: `Q_Levels_Database.csv`
- Frequency: quarterly ($T = 160$ obs)

(2) Variables

Code	Definition	Role	Form	Source
LGPR	World GPR index	shock target	log	TBD
LGPRT	GP threats sub-index	shock target	log	TBD
LGPRa	GP acts sub-index	shock target	log	TBD
Unemp	US unemployment rate	control (t)	rate	TBD
FFR	federal funds rate	control (t)	rate	TBD
Real_Brent	real Brent price	control ($t-1$) + y	real (CPI-defl.)	TBD
Real_WTI	real WTI price	control ($t-1$) + y	real (CPI-defl.)	TBD
Real_Gasoline	real gasoline price	control ($t-1$) + y	real (CPI-defl.)	TBD
Headline_Pi	headline CPI	LP outcome (y)	CPI inflation	TBD
Core_Pi	core CPI	LP outcome (y)	CPI inflation	TBD
Indu_Prod	industrial production	LP outcome (y)	TBD	TBD

Note: To be double-checked!

- **LGPR, LGPRT, LGPRa:** codebook says “log”, but 1986Q1 value ≈ 470 matches $100 \times \ln(\text{GPR})$ for raw GPR ≈ 110 (not $\ln(\text{GPR}) \approx 4.7$). Likely $100 \times \log$ in practice.
- **Headline_Pi, Core_Pi:** codebook says “CPI inflation”, but 1986Q1 value ≈ 470 matches $100 \times \ln(\text{CPI level})$, not an inflation rate (which would be 1–5%).

(3) Sample window

All available observations are used. **The binding constraint is the late start of Real_Brent (first non-missing 1987Q3).**

Stage	n
Raw quarterly data (1986Q1–2025Q4)	160
After $\text{AR}(p = 4)$ lag truncation	156
Shock sample, w/ oil	153
Shock sample, w/o oil	156
LP sample at $h = 0$, $y \in \{\text{WTI}, \text{Gas}\}$	153
LP sample at $h = 0$, $y \in \{\text{Brent}, \text{macro}\}$	150
LP sample at horizon h	(above) $-h$

Note:

- w/ oil loses 3 extra obs: needs lag-1 Brent at $t = 5, 6, 7$, missing.
- LP needing Brent at $t - 1, \dots, t - 4$ starts at $t = 11$ (1988Q3) $\Rightarrow$ 150 rows.
- LP at horizon h loses h end-of-sample rows ($t + h \leq T$).

(4) Oil-price multicollinearity

Pairwise correlations of the three lag-1 oil controls:

	Real_Brent	Real_WTI	Real_Gasoline
Real_Brent	1.000	0.993	0.976
Real_WTI	0.993	1.000	0.973
Real_Gasoline	0.976	0.973	1.000

$\Rightarrow$ Correlations of 0.97–0.99 flag substantial multicollinearity.

3 GPR Shock Extraction**(1) AR specification**

For each GPR series the standardized shock z_t is the OLS residual of

$$\text{GPR}_t = a + \sum_{j=1}^p \rho_j \text{GPR}_{t-j} + \xi' B_t + \omega' O_{t-1} + u_t, \quad z_t = \frac{u_t - \bar{u}}{s_u}, \quad (1)$$

with $p = 4$, $B_t = [\text{Unemp}_t, \text{FFR}_t]$, and $O_{t-1} = [\text{Brent}, \text{WTI}, \text{Gas}]_{t-1}$.

(2) Identifying logic for control timing

Controls	Timing	Reason
Unemp, FFR	contemporaneous (t)	slow-moving; no bad-control bias
Brent, WTI, Gas	lag-1 ($t - 1$)	contemporaneous use would partial out the transmission channel

(3) Shock-extraction summary

Variant	Shock	nobs	AR R^2	Resid sd	cond($X'X$)
w/ oil	World	153	0.507	22.96	5.6×10^9
w/ oil	GPT	153	0.515	22.58	5.8×10^9
w/ oil	GPA	153	0.538	30.44	5.4×10^9
w/o oil	World	156	0.505	22.79	2.9×10^8
w/o oil	GPT	156	0.512	22.45	3.2×10^8
w/o oil	GPA	156	0.529	30.43	1.4×10^8

(4) Takeaways

- **Specification robust:** $\Delta R^2 < 0.01$, $\Delta \text{sd} < 0.2$ across variants.
- **Two shocks essentially identical.**
- **Oil controls inflate $\text{cond}(X'X)$ by $\sim 20\times$,** still well below 10^{12} .
- **Multicollinearity is benign for the residual:** residual depends on the column space, not on individual columns.

4 Shock Validation Diagnostics

(1) Test 1 — Conditional Granger exogeneity

- H_0 : lagged X_j does not Granger-cause z_t , conditional on lags of the other four predictors in $\{\text{Unemp}, \text{FFR}, \text{Real_Brent}, \text{Real_WTI}, \text{Real_Gasoline}\}$.
- Tool: `gctest` with $L = 4$.
- **Results** (30 tests = 2 variants $\times$ 3 shocks $\times$ 5 predictors):

Variant	Rejections at $\alpha = 0.05$	Smallest p
w/ oil	1 / 15 (FFR $\rightarrow z_{\text{GPA}}$, $p = 0.032$)	0.032
w/o oil	0 / 15	0.065
Combined	1 / 30	0.032

- 28 of 30 p -values exceed 0.10.
- Expected false positives at $\alpha = 0.05$: $1.5 \Rightarrow 1$ rejection within chance.
- **Bonferroni ($\alpha = 0.05/30 \approx 0.0017$): 0 rejections.**
- **Conclusion: shocks are exogenous to past macro/oil-price information.**

(2) Test 2 — Ljung-Box Q (white noise)

- H_0 : z_t is white noise (no serial correlation up to lag L).
- Tool: `lbqtest`.
- **Results** (18 tests = $2 \times 3 \times 3$):

Rejections	Min p -value	Conclusion
0 / 18	0.798	no serial correlation

(3) Test 3 — Breusch-Godfrey LM (robustness)

- H_0 : $\gamma_1 = \dots = \gamma_L = 0$ in

$$z_t = X_t' \alpha + \sum_{k=1}^L \gamma_k z_{t-k} + v_t, \quad \text{LM} = nR_{\text{aux}}^2 \xrightarrow{d} \chi^2(L).$$

X_t = original $\text{AR}(p)$ regressors.

- Preferred over LBQ when lagged dependent variables enter the regression (our case).
- **Results** (18 tests):

Rejections	Min LM p -value	Conclusion
0 / 18	0.319	no serial correlation (robust)

- BG p -values are systematically lower than LBQ (BG partials out the original regressors) but lead to the same conclusion.

(4) Summary of diagnostics

Test	Rejections	Min p	Conclusion
1. Conditional Granger	1 / 30 (borderline)	0.032	shocks exogenous
2. Ljung-Box Q	0 / 18	0.798	no serial correlation
3. Breusch-Godfrey LM	0 / 18	0.319	confirmed (robust)

(5) Face validity — event alignment of z_t

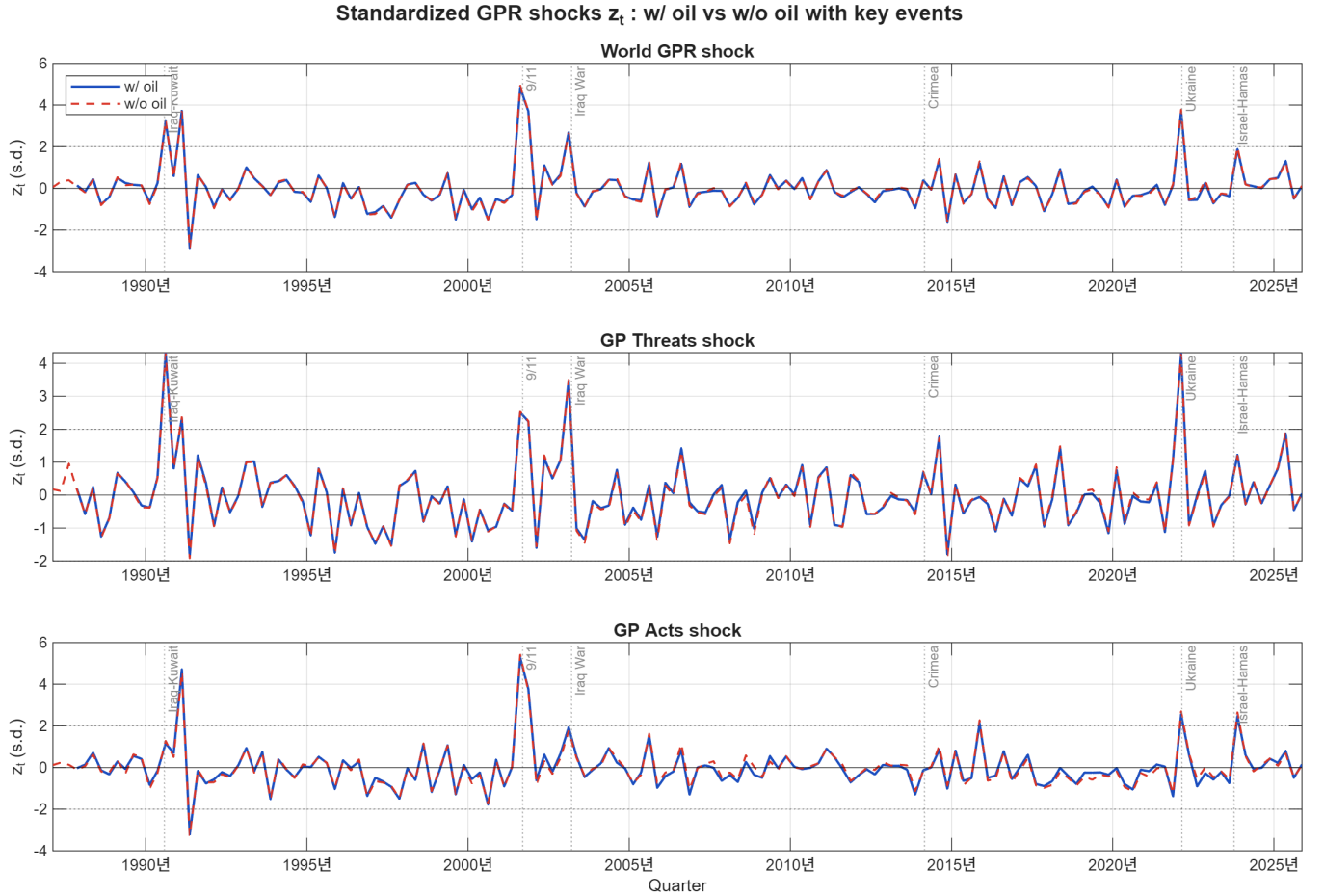


Figure 1: Standardized GPR shocks z_t over 1986Q1–2025Q4

⇒ **Spike alignment with major events** (w/ oil variant z_t values):

Event	Quarter	World	GPT (threats)	GPA (acts)
Gulf War (Iraq–Kuwait)	1990Q3	+3.23	+4.32	+1.16
9/11	2001Q3	+4.83	+2.51	+5.27
Iraq War	2003Q1	+2.69	+3.48	+1.93
Crimea annexation	2014Q1	+0.39	+0.68	−0.13
Russia–Ukraine	2022Q1	+3.70	+4.29	+2.54
Israel–Hamas	2023Q4	+1.88	+1.22	+2.46

(Sample max in bold per column.)

⇒ **Observations:**

- All major events except Crimea produce positive z_t spikes; magnitudes broadly scale with event size.
- w/ oil (blue) and w/o oil (red) lines are visually indistinguishable (the spec-robustness claim).
- **Sample maxima:** 9/11 dominates World and GPA (4.83 and 5.27); Gulf War dominates GPT (4.32, with Russia–Ukraine 4.29 essentially tied).
- The 2014 Crimea annexation barely registers, possibly because the annexation was a contained event with limited global escalation at the time.

⇒ **Conclusion:** z_t is not only statistically white noise but also carries **economically meaningful information** about geopolitical events.

5 Baseline LP Model

(1) Specification

For each shock z_t and outcome y , at horizon $h = 0, 1, \dots, H$:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h z_t + \sum_{j=1}^p \rho_{h,j} y_{t-j} + \sum_{j=1}^p \gamma_{h,j} \text{GPR}_{t-j} + \sum_{j=1}^p \theta'_{h,j} \text{controls}_{t-j} + e_{t+h}. \quad (2)$$

(2) Settings

Setting	Value
Lag length	$p = 4$ (matches AR shock extraction)
Horizon	$h = 0, \dots, H = 12$ quarters
Shock size	1 s.d. (since z_t is standardized)
Inference	Newey–West HAC, Bartlett kernel, bandwidth = $h + 1$
Confidence band	90%

(3) Outcome-dependent control set

To avoid SE inflation from oil-price multicollinearity, the full control pool is pruned dynamically based on the outcome variable:

Outcome (y)	Control Variables
Oil price (any of Real_Brent/WTI/Gasoline)	Unemp, FFR, Headline_Pi, Core_Pi, Indu_Prod
Headline_Pi	Unemp, FFR, Real_Brent, Real_WTI, Real_Gasoline, Core_Pi, Indu_Prod
Core_Pi	Unemp, FFR, Real_Brent, Real_WTI, Real_Gasoline, Headline_Pi, Indu_Prod
Indu_Prod	Unemp, FFR, Real_Brent, Real_WTI, Real_Gasoline, Headline_Pi, Core_Pi

6 Results

(1) Shock-specification robustness

Across all 18 panels (3 shocks $\times$ 6 outcomes), w/ oil and w/o oil IRFs and bands are visually indistinguishable (see Section 6-(6) for IRF figures).

(2) Oil-price peaks

Peak responses in the w/ oil variant (1 s.d. shock; positive = price rise; $H = 12$):

Shock	Outcome	Peak			β_0	
		β	h	t	β	t
GPT	Brent	+7.16	7	3.27	+2.66	1.73
GPT	WTI	+6.49	7	3.17	+2.40	1.73
GPT	Gasoline	+3.11	7	2.61	+0.59	0.92
World	Brent	+4.55	9	2.24	+1.23	0.74
World	WTI	+4.15	9	2.27	+0.94	0.61
World	Gasoline	+2.16	9	1.95	-0.14	-0.19
GPA	Brent	+3.82	9	2.19	-0.75	-0.61
GPA	WTI	+3.75	12	1.78	-0.71	-0.60
GPA	Gasoline	+1.97	9	2.11	-0.75	-1.14

- **GPT (threats)**: short-horizon hump, peaks at $h = 7$. Strongest significance ($t \approx 3$).
- **World, GPA (acts)**: delayed peaks at $h \approx 9$ –12, smaller magnitudes than GPT.
- **Peak significance**: 8 of 9 peaks clear 90% ($|t| \geq 1.645$); only GPA→WTI at $h = 12$ is borderline ($t = 1.78$).

- w/ oil vs w/o oil differ by < 0.5 at peak (shock-spec robust).

Caveat (pointwise significance across horizons). The peak rows above pick the single most-significant horizon. Across all $h = 0, \dots, 12$, the share of horizons clearing the 90% band varies considerably:

Shock	Brent	WTI	Gasoline
GPT	7 / 13 (54%)	7 / 13 (54%)	5 / 13 (38%)
World	5 / 13 (38%)	7 / 13 (54%)	4 / 13 (31%)
GPA	1 / 13 (8%)	3 / 13 (23%)	4 / 13 (31%)

$\Rightarrow$ The peak finding is real (at least one horizon is significant for every cell), but **persistence across horizons is uneven**, especially for GPA. Visual IRFs in Section 6-(6) show this directly: bands often contain zero outside a short window around the peak.

(3) Inflation peaks

$y_{t+h} - y_{t-1}$ = cumulative % change in CPI. Most-negative point estimates ($H = 12$):

Shock	Outcome	β	h	t
World	Headline_Pi	-0.131	10	-0.99
World	Core_Pi	-0.143	12	-1.28
GPT	Headline_Pi	-0.089	12	-0.52
GPT	Core_Pi	-0.133	12	-0.97
GPA	Headline_Pi	-0.199	10	-1.71
GPA	Core_Pi	-0.179	10	-1.93

Pointwise significance across horizons ($h = 0, \dots, 12$):

Shock	Headline_Pi	Core_Pi
World	0 / 13 (0%)	0 / 13 (0%)
GPT	0 / 13 (0%)	0 / 13 (0%)
GPA	4 / 13 (31%)	6 / 13 (46%)

- Inflation responses to GPR shocks are largely **statistically indistinguishable from zero**.
- **4 of 6 cells have 0 / 13 significant horizons** (World and GPT for both Headline and Core). The entire IRF stays within the 90% band of zero.
- Only GPA $\rightarrow$ Headline/Core produce borderline signal (peak $t \approx -1.7$ to -1.9 , 4–6 of 13 horizons significant), in the **negative** (deflationary) direction.
- Contemporaneous response ≈ 0 everywhere: $|\beta_0| \leq 0.06$ (headline), $|\beta_0| \leq 0.02$ (core).
- The slide-deck’s cost-push prior (“threats inflate, acts deflate”) is **not supported**: GPT shows no effect, and GPA points in the deflationary direction for both Headline and Core.

- **Interpretation:** Either (i) the demand channel (uncertainty $\rightarrow$ contraction $\rightarrow$ disinflation) dominates the cost-push channel; or (ii) the macro block in the LP control set absorbs the cost-push channel through lagged oil prices, leaving only the residual demand effect; or (iii) the GPR-to-inflation spillover is too small relative to other macro noise at the quarterly frequency to be detected with 153 observations.

(4) Industrial production peaks

Most-significant horizon per shock (w/ oil variant; $H = 12$):

Shock	β	h	t	Note
GPT	-0.007	11	-1.50	-0.7%, borderline ($p \approx 0.13$)
World	-0.003	11	-0.81	not significant
GPA	-0.003	1	-1.80	-0.3%, sig at 90% (short-run only)

Pointwise significance across horizons ($h = 0, \dots, 12$):

Shock	Indu_Prod
World	0 / 13 (0%)
GPT	1 / 13 (8%)
GPA	1 / 13 (8%)

- IP responses are **even weaker than inflation** — the IRFs stay near zero with wide bands across almost all horizons.
- The same three explanations as for inflation apply, plus a fourth: IP is even slower-moving than inflation, so a 12-quarter horizon may simply be too short to detect macro spillover from GPR shocks.

LP IRF comparison: World GPR shock (90% bands, NW HAC bw=h+1)

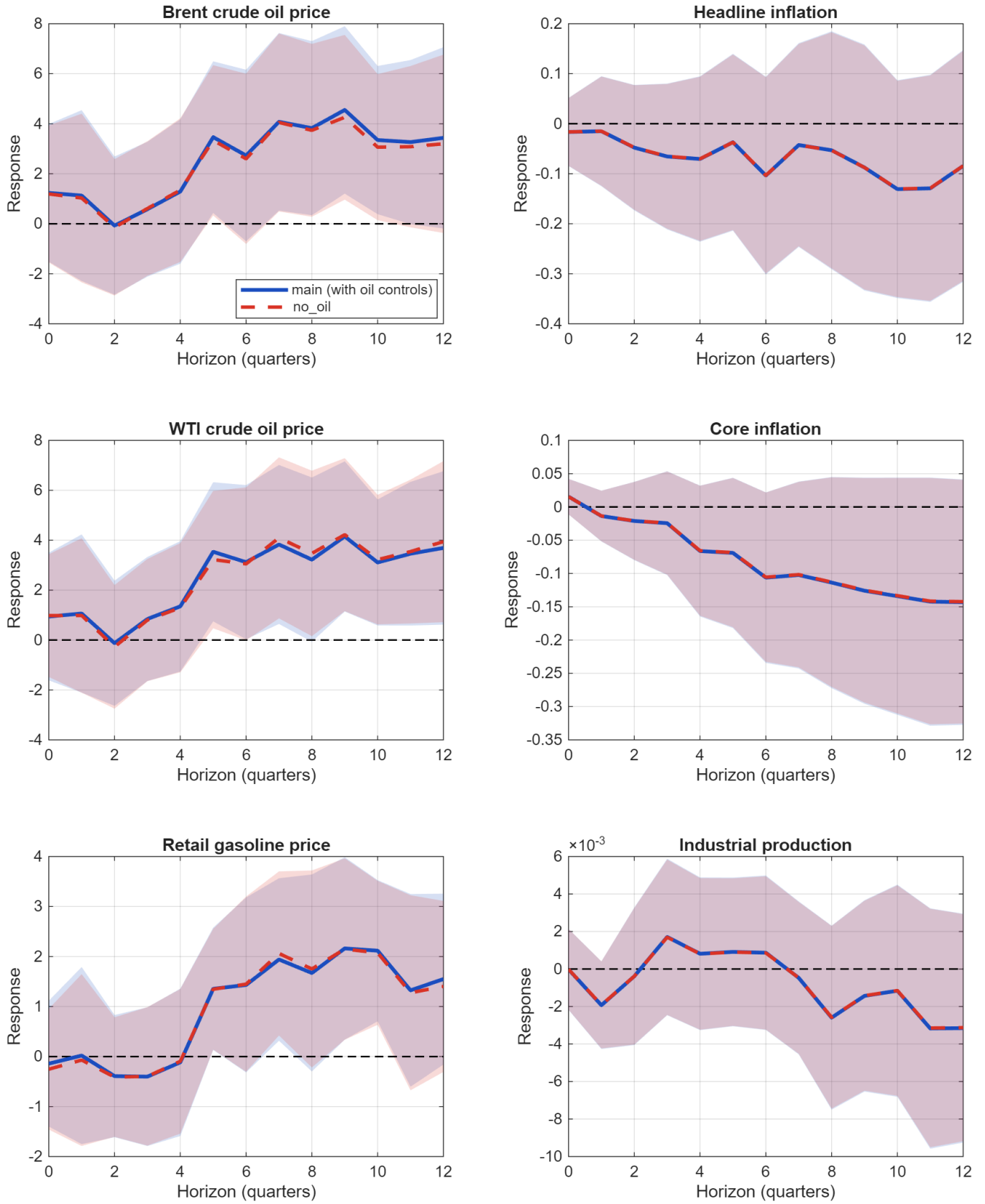


Figure 2: IRF to a 1 s.d. World GPR shock. Shaded: 90% NW HAC bands, bandwidth $h + 1$.

LP IRF comparison: GPT GPR shock (90% bands, NW HAC bw=h+1)

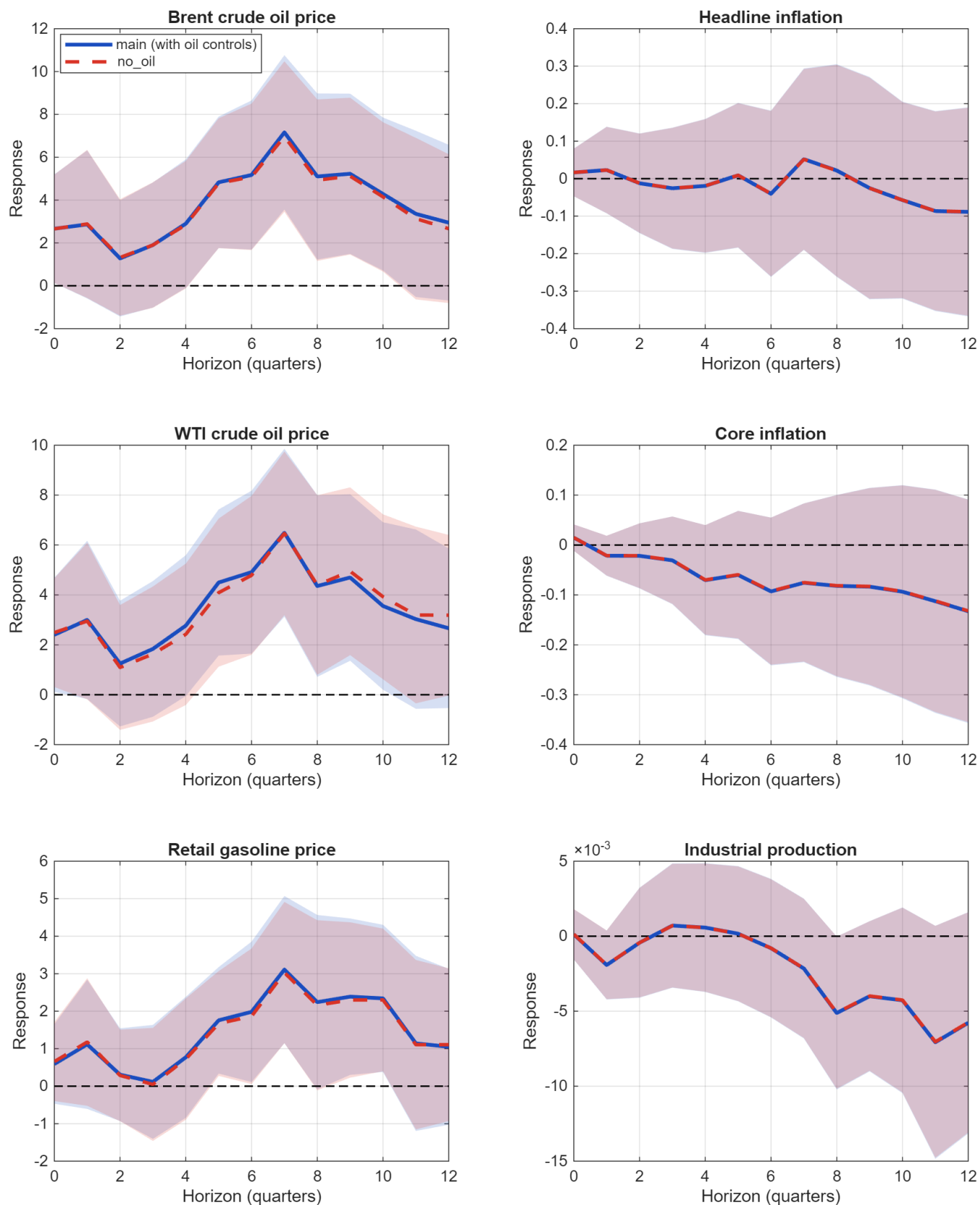


Figure 3: IRF to a 1 s.d. GPT (threats) shock. Conventions as above.

LP IRF comparison: GPA GPR shock (90% bands, NW HAC bw=h+1)

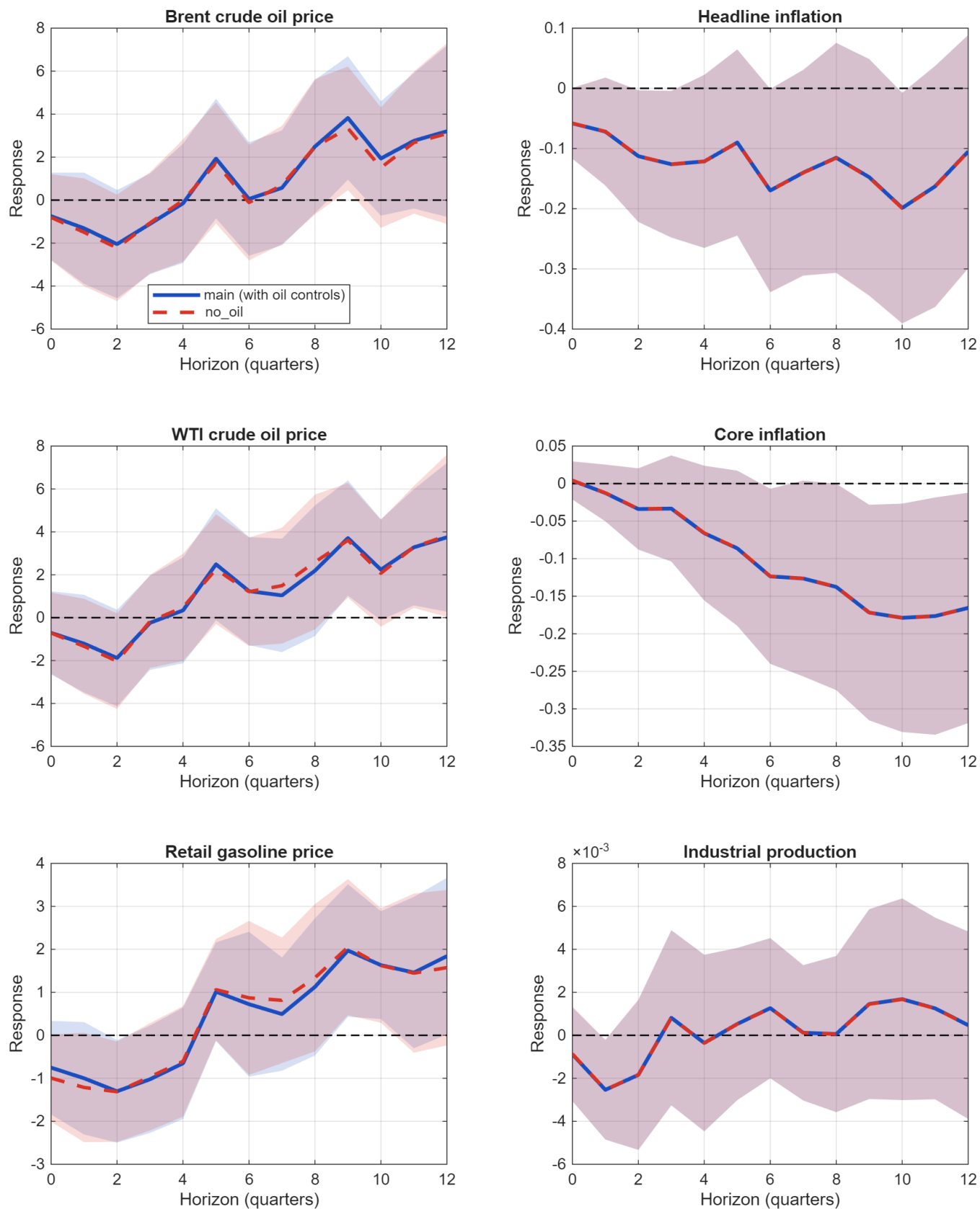


Figure 4: IRF to a 1 s.d. GPA (acts) shock. Conventions as above.